

PV250HE / PV375HE / PV500HE

Communication Protocol



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1 Introduction

This document intends to be a guide for implementation of remote monitoring for the PV250E/PV375HE/PV500HE Photovoltaic Inverters.

These equipments are equipped with a communication interface that can be accessed using the MODBUS protocol.

There are available two options, MODBUS RTU or MODBUS TCP. For more information about the hardware and cable connection, please consult the equipment installation and operation manuals.

For more information about the MODBUS protocol please consult the website <http://www.modbus.org/>.

2 Configuration

The Photovoltaic Inverter is configured as a Slave node in the communication network and can be accessed by MODBUS RTU or MODBUS TCP.

The Register Map is also described in the next sections and can be easily implemented on the MODBUS Master controller.

2.1 MODBUS RTU

In RTU (Remote Terminal Unit) Mode, each 8-bit message byte contains two 4-bit hexadecimal characters, and the message is transmitted in a continuous stream via a serial port. The settings for this interface are listed below:

Mode:	RTU
Baudrate:	19200
Parity:	Even
Databits:	8 bits

The Slave ID can be configured easily on the HMI Interface, in the menu option MODBUS.

Other settings can be configured by Efacec personnel, if requested. The MODBUS RTU is available on a RS485 half duplex port. For more information about the details of this port, consult the equipment installation manual.

2.2 MODBUS TCP

MODBUS TCP/IP uses TCP/IP and Ethernet to carry the MODBUS messaging structure. On this mode, it should be defined the network settings (IP address and port) and the Slave ID.

This configuration can be done on the Photovoltaic Inverter HMI Interface in the menu MODBUS.

3 Register Map

The Photovoltaic Inverter information is mapped in Holding Registers, divided in five sections, as follows:

- 40001 to 40040 – Global Information
- 40041 to 40086 – Inverter 1 information
- 40087 to 40132 – Inverter 2 information
- 40133 to 40178 – Inverter 3 information
- 40179 to 40223 – Inverter 4 information

Please note that the values labeled as float in the Var Type are encoded as single precision IEEE-754 big endian.

Addr	Var Name	Var Type		Reg Idx	Inverter Addr	Units	Comments
40001 40002	P	float	High Low	r	Global	kW	Global Active Power
40003 40004	Q	float	High Low	r	Global	kVA	Global Reactive Power
40005 40006 ... 40040	Comm. Status	uint16		r	Global Global	-	Communication Error Status – Consult Table 2 Cos Phi table – Consult Table 3
40041 ... 40086					1		Inverter 1 Information - Consult Table 4
40087 ... 40132					2		Inverter 2 Information - Consult Table 4
40133 ... 40178					3		Inverter 3 Information - Consult Table 4
40179 ... 40223					4		Inverter 4 Information - Consult Table 4

Table 1 – Register Map

The next tables present Communication Error Status. Please note that the values presented on these fields can be combined when more than one error or condition is active. In the following tables the Mask should be used with a bitwise AND operation to query the status of the bit.

Mask	Comment
0x0001	Communication error in Inverter 1 (1 – error, 0 - normal)
0x0002	Communication error in Inverter 2 (1 – error, 0 - normal)
0x0004	Communication error in Inverter 3 (1 – error, 0 - normal)
0x0008	Communication error in Inverter 4 (1 – error, 0 - normal)

Table 2 – Communication Error Status

Offset	Var Name	Var Type	Reg Idx	Units	Comments
0	Summer_init_day	uint16	r		Day of the beginning of summer for cos phi control
1	Summer_init_month	uint16	r		Month of the beginning of summer
2	S_peak_init_hour	uint16	r	h	
3	S_peak_init_min	uint16	r	min	
4	S_peak_end_hour	uint16	r	h	
5	S_peak_end_min	uint16	r	min	
6	S_peak_cosphi	int16	r		
7	S_valley_init_hour	uint16	r	h	
8	S_valley_init_min	uint16	r	min	
9	S_valley_end_hour	uint16	r	h	
10	S_valley_end_min	uint16	r	min	
11	S_valley_cosphi	int16	r		
12	S_flat_init_hour	uint16	r	h	
13	S_flat_init_min	uint16	r	min	
14	S_flat_end_hour	uint16	r	h	
15	S_flat_end_min	uint16	r	min	
16	S_flat_cosphi	int16	r		
17	Winter_init_day	uint16	r	h	Day of the beginning of winter for cos phi control
18	Winter_init_month	uint16	r	h	Month of the beginning of winter
19	W_peak_init_hour	uint16	r	h	
20	W_peak_init_min	uint16	r	min	
21	W_peak_end_hour	uint16	r	h	
22	W_peak_end_min	uint16	r	min	
23	W_peak_cosphi	int16	r		
24	W_valley_init_hour	uint16	r	h	
25	W_valley_init_min	uint16	r	min	
26	W_valley_end_hour	uint16	r	h	
27	W_valley_end_min	uint16	r	min	
28	W_valley_cosphi	int16	r		
29	W_flat_init_hour	uint16	r	h	
30	W_flat_init_min	uint16	r	min	
31	W_flat_end_hour	uint16	r	h	
32	W_flat_end_min	uint16	r	min	
33	W_flat_cosphi	int16	r		

Table 3 – CosPhi table

Offset	Var Name	Var Type	Reg Idx	Units	Comments
0	Vdc	float	High	V	Input voltage
1		Low	r		
2	Idc	float	High	A	Input current
3		Low	r		
4	Pdc	float	High	kW	Input Power
5		Low	r		
6	Vrs	float	High	V	Output Voltage (Vrs)
7		Low	r		
8	Vst	float	High	V	Output Voltage (Vst)
9		Low	r		
10	Vtr	float	High	V	Output Voltage (Vtr)
11		Low	r		
12	Ir	float	High	A	Output Current (Ir)
13		Low	r		
14	Is	float	High	A	Output Current (Is)
15		Low	r		
16	It	float	High	A	Output Current (It)
17		Low	r		
18	P	float	High	kW	Output Power
19		Low	r		
20	FP	float	High		Power Factor
21		Low	r		
22	Temp	float	High	°C	Temperature
23		Low	r		
24	Freq	float	High	Hz	Frequency
25		Low	r		
26	Alrm_Vgrid	uint16		r	0 - Normal, X - Grid Voltage Alarm (*)
27	Alrm_Freq	uint16		r	0 - Normal, X - Grid Frequency Alarm (*)
28	Alrm_Fus	uint16		r	0 - Normal, X - Surge Arresters Alarm (*)
29	Alrm_Fearth	uint16		r	0 - Normal, X - Earth Fault Alarm (*)
30	Alrm_Prot_Int	uint16		r	0 - Normal, X - Internal Protections Alarm (*)
31	Alrm_OverTemp	uint16		r	0 - Normal, X - Overtemperature Alarm (*)
32	PreAlrm_OverTemp	uint16		r	0 - Normal, X - Overtemperature Warning (*)
33	Alrm_Fans	uint16		r	0 - Normal, X - Fans Failure Alarm (*)
34	CounterP	float	High	kWh	Active Energy Counter
35		Low	r		
36	CounterQ	float	High	kVAh	Reactive Energy Counter
37		Low	r		
38	CosPhi	int16		r	Reserved
39	Failure	uint16		r	0 - Normal, 1 - Inverter Failure
40	FlagErr1	uint32	High	r	Internal Error Flag 1 – Consult Table 5
41		Low	r		
42	FlagErr2	uint32	High	r	Internal Error Flag 2- Consult Table 6
43		Low	r		
44	Command	uint16		r/w	Reserved
45	MasterSlave	uint16		r	Master/Slave Status – Consult Table 7

Table 4 – Inverter module Information

(*) – X: other than zero.

The next tables present Communication Error Status. Please note that the values presented on these fields can be combined when more than one error or condition is active.

Mask	Comment
0x00000001	MASK_ERR_VGRID_HIGH
0x00000002	MASK_ERR_VGRID_LOW
0x00000004	MASK_ERR_FREQ_HIGH
0x00000008	MASK_ERR_FREQ_LOW
0x00000010	MASK_ERR_VGRID_PEAK
0x00000020	MASK_ERR_IGRID_PEAK
0x00000040	MASK_ERR_IDC_PEAK
0x00000080	MASK_ERR_OVERLOAD
0x00000100	MASK_ERR_OVERTEMP
0x00000200	MASK_ERR_PREC_VPV_LOW
0x00000400	MASK_ERR_PREC_VCAP_LOW
0x00000800	MASK_ERR_VPV_HIGH
0x00001000	MASK_ERR_VPV_LOW
0x00002000	MASK_ERR_VCAP_HIGH
0x00004000	MASK_ERR_VCAP_LOW
0x00008000	MASK_ERR_DRIVER
0x00010000	MASK_ERR_RESET_DSP
0x00020000	MASK_ERR_KP_INPUT
0x00040000	MASK_ERR_KCC_INPUT
0x00080000	MASK_ERR_KS_INPUT
0x00100000	MASK_ERR_FUS_VARISTOR
0x00200000	MASK_ERR_EARTH_FAULT
0x00400000	MASK_ERR_INT_ERROR
0x00800000	MASK_ERR_KI_INPUT
0x01000000	MASK_ERR_KVENT_INPUT
0x02000000	MASK_ERR_IDC_LOW
0x04000000	MASK_ERR_POWER_LOW
0x08000000	MASK_ERR_I_ASYM
0x10000000	MASK_ERR_V_ASYM
0x20000000	MASK_ERR_KT_INPUT
0x40000000	MASK_ERR_OUTPUT_CHECK
0x80000000	MASK_ERR_KCONTROL

Table 5 – Internal Error Flag 1

Mask	Comment
0x00000001	MASK_ERR_EARTH_FAULT_TN
0x00000002	MASK_ERR_EARTH_FAULT_TS
0x00000004	MASK_ERR_UPS_INPUT
0x00000008	MASK_ERR_UPS_OUTPUT
0x00000010	MASK_ERR_FUS_VARISTOR_AC
0x00000020	MASK_ERR_VARISTOR_DC
0x00000040	MASK_ERR_KVENT_STACK
0x00000080	MASK_ERR_KVENT_COIL
0x00000100	MASK_ERR_DRIVER_R
0x00000200	MASK_ERR_DRIVER_S
0x00000400	MASK_ERR_DRIVER_T
0x00000800	MASK_ERR_VENT_CB
0x00001000	MASK_ERR_KI_INPUT
0x00002000	MASK_ERR_PRECHARGE_AC
0x00004000	MASK_ERR_HOMOPOLAR_VOLTAGE
0x00008000	MASK_ERR_2_15
0x00010000	MASK_ERR_2_16
0x00020000	MASK_ERR_2_17
0x00040000	MASK_ERR_2_18
0x00080000	MASK_ERR_2_19
0x00100000	MASK_ERR_2_20
0x00200000	MASK_ERR_2_21
0x00400000	MASK_ERR_2_22
0x00800000	MASK_ERR_2_23
0x01000000	MASK_ERR_2_24
0x02000000	MASK_ERR_2_25
0x04000000	MASK_ERR_2_26
0x08000000	MASK_ERR_2_27
0x10000000	MASK_ERR_2_28
0x20000000	MASK_ERR_2_29
0x40000000	MASK_ERR_2_30
0x80000000	MASK_ERR_2_31

Table 6 – Internal Error Flag 2

Mask	Comment
0x0003	Master/Slave mode: 00 – Slave 01 – Master Type I 10 – Master Type II
0x0004	Inverter Status: 0 – Off 1 – On
0x0008	Interconnection DC switch status: 0 – Open 1 – Closed

Table 7 – Master / Slave status

4 Remote Monitoring System

The Register Map described on this document is complete list of the information available on the Inverter Communication Interface.

It is advisable to consult EFACEC to discuss the implementation of the Remote Monitoring System and selection of the most important data for this purpose.